

Pattern of acute coronary syndrome presentation in women

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Abstract

Introduction: Cardiovascular disease (CVD) constitutes the leading cause of death for women in developed countries. It is a more common cause of death than cancer and leads to one death every minute. Results of our study helps in understanding the pattern of acute coronary syndrome presentation in women. **Discussion and Conclusion:** Unstable angina and anterior wall myocardial infarction formed almost 60% of our study group. Effort angina was only seen in 5.3% of patient. AWMI and unstable angina was the commonest presentation across all age groups. Hypertension was seen in 60% of patients presenting with ACS and mode of pattern did not differ significantly. AWMI and unstable angina was the most common presentation in diabetic patients

Keywords: ACS: acute coronary syndrome, CAD: Coronary artery disease, AWMI: anterior wall myocardial infarction, IWMI: inferior wall mi, UA: unstable angina, EA: effort angina, NSTEMI: non ST elevation mi.

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INTRODUCTION

Coronary heart disease is the leading cause of mortality and morbidity in men as well as in women. The type of ischemic event shows gender-specific differences. Studies such as GUSTO IIb (Global Use of Strategies to Open Occluded Coronary Arteries in Acute Coronary Syndromes),¹ TIMI IIIB (Thrombolysis In Myocardial Infarction),² and the Euro Heart Survey,³ women present more frequently with unstable angina and non-ST-elevation myocardial infarction (NSTEMI), whereas men have ACS with ST elevation (STEMI).

Research gap

Results of our study helps in understanding the pattern of acute coronary syndrome presentation in women.

OBJECTIVES

Primary objective

To determine pattern of acute coronary syndrome presentation in women.

MATERIAL AND METHODS

Study Design

Retrospective descriptive observational study. Study setting and population Patients diagnosed with acute coronary syndrome admitted between December 2012 to Jan 2015 to Sri Jayadeva institute of cardiovascular sciences and research, Bangalore and Bangalore medical college and research institute Bangalore formed our study group. The period of study was from December 2012 to Jan 2015.

Data Requirement

The following data were studied by reviewing records entered in case records and comic computer data: age, sex, risk factors for coronary artery disease were obtained from the details entered at the time of diagnoses. Data collected was maintained in strict confidentiality with access to Principal Investigator and authorized study personnel.

Inclusion Criteria

All women patient presenting with ischemic heart disease satisfying the below criteria were included in our study.

The diagnosis of myocardial infarction was based on least two of the following: characteristic chest pain, a cardiac enzyme rise greater than twice the upper normal limit, and compatible ECG changes lasting >48 hours or demonstration of regional wall motion abnormality with imaging modalities.

Exclusion Criteria

1. Patients presenting for second time were excluded from study.
2. With underlying neoplasm and retroviral disease were excluded.

Statistical Analysis

Method of statistical analysis: Descriptive analysis was carried out in this study. The results were expressed as mean±SD. Significance was assessed to 5% level of significance. Software namely SPSS 15 was used for the analysis of the data. Microsoft word and Excel were used to generate graphs and tables.

Ethical issues

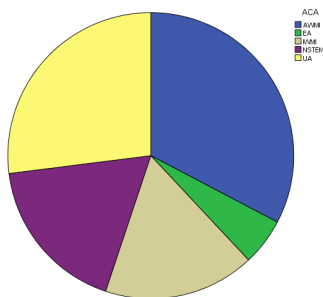
The study involved mainly observational. No incentive was obtained for the study in any form. Authors don't have any conflict of interest.

RESULTS

The study group consisted of 263 patients. The age of the patients ranged between 30-83 years.

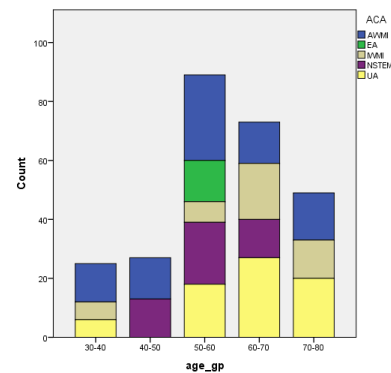
Table 1: Pattern of acute coronary syndrome presentation in women

	Frequency	Percent	Valid Percent
AWMI	86	32.7	32.7
EA	14	5.3	5.3
IWMI	45	17.1	17.1
NSTEMI	47	17.9	17.9
UA	71	27.0	27.0
Total	263	100.0	100.0



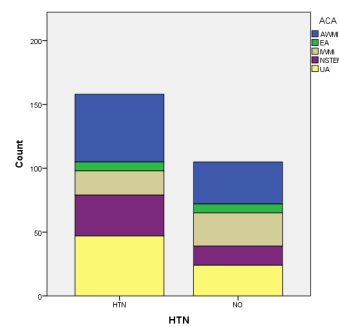
Anterior wall myocardial infarction (AWMI) was the commonest presentation in 33% of patients; unstable angina (UA) was the next with 27%. Unstable angina and anterior wall myocardial infarction formed almost 60% of our study group. Inferior wall myocardial infarction and NSTEMI were almost equal with 17%. Effort angina was only seen in 5.3% of patient.

Age and pattern of ACS presentation in women



Effort angina was seen in 50-60 years of age. NSTEMI was not seen at extremes of age. Unstable angina was equally distributed among all the age groups. AWMI and unstable angina was the commonest presentation across all age groups.

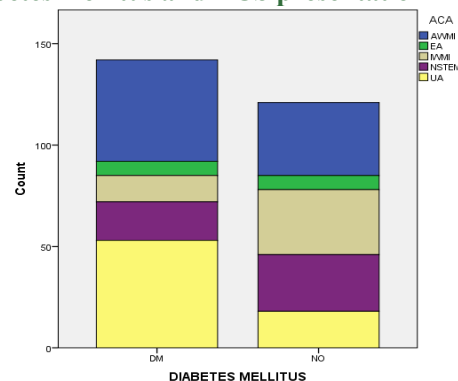
Hypertension and ACS presentation in women



HTN					
	Frequency	Percent	Valid Percent	Cumulative Percent	
Valid HTN	158	60.1	60.1	60.1	
Valid NO	105	39.9	39.9	100.0	
Total	263	100.0	100.0		

Hypertension was seen in 60% of patients presenting with ACS and mode of pattern did not differ significantly.

Diabetes mellitus and ACS presentation in women



Diabetes Mellitus				
	Frequency	Percent	Valid Percent	Cumulative Percent
DM	142	54.0	54.0	54.0
NO	121	46.0	46.0	100.0
Total	263	100.0	100.0	

Diabetes mellitus was seen in 54% patient. Unstable angina and AAMI formed more than 70% of the ACS group.

DISCUSSION

Unstable angina and anterior wall myocardial infarction formed almost 60% of our study group. Effort angina was only seen in 5.3% of patient. AAMI and unstable angina was the commonest presentation across all age groups. Hypertension was seen in 60% of patients presenting with ACS and mode of pattern did not differ significantly. AAMI and unstable angina was the most common presentation in diabetic patients.

CONCLUSION

Unstable angina and anterior wall myocardial infarction is the commonest mode of presentation of ACS in women across all age groups in both diabetic and non diabetic women. NSTEMI was uncommon in extremes of age.

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